
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Asaba, Joshua
Date of birth:	2/24/1990
Subject:	050-Meteorology
Result:	27,5 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	365	50.	METEOROLOGY	0,00	1,00
2	167	50.	METEOROLOGY	0,00	1,00
3	276	50.	METEOROLOGY	0,00	1,00
4	440	50.	METEOROLOGY	0,00	1,00
5	210	50.	METEOROLOGY	1,00	1,00
6	341	50.	METEOROLOGY	1,00	1,00
7	318	50.	METEOROLOGY	0,00	1,00
8	351	50.	METEOROLOGY	0,00	1,00
9	438	50.	METEOROLOGY	0,00	1,00
10	355	50.	METEOROLOGY	0,00	1,00
11	540	50.	METEOROLOGY	0,00	1,00
12	451	50.	METEOROLOGY	0,00	1,00
13	433	50.	METEOROLOGY	1,00	1,00
14	273	50.	METEOROLOGY	0,00	1,00
15	316	50.	METEOROLOGY	0,00	1,00
16	174	50.	METEOROLOGY	1,00	1,00
17	306	50.	METEOROLOGY	0,00	1,00
18	195	50.	METEOROLOGY	1,00	1,00
19	258	50.	METEOROLOGY	0,00	1,00
20	503	50.	METEOROLOGY	0,00	1,00
21	533	50.	METEOROLOGY	0,00	1,00
22	337	50.	METEOROLOGY	0,00	1,00
23	285	50.	METEOROLOGY	1,00	1,00
24	537	50.	METEOROLOGY	0,00	1,00
25	542	50.	METEOROLOGY	0,00	1,00
26	264	50.	METEOROLOGY	0,00	1,00
27	360	50.	METEOROLOGY	0,00	1,00
28	350	50.	METEOROLOGY	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	357	50. METEOROLOGY		1,00	1,00
30	272	50. METEOROLOGY		0,00	1,00
31	414	50. METEOROLOGY		0,00	1,00
32	504	50. METEOROLOGY		0,00	1,00
33	474	50. METEOROLOGY		0,00	1,00
34	219	50. METEOROLOGY		1,00	1,00
35	27	50.2.2.1	Primary cause of wind, pressure gradient,	1,00	1,00
36	81	50.2.4.1	Anabatic and catabatic winds etc	1,00	1,00
37	310	50.3	THERMODYNAMICS	0,00	1,00
38	125	50.4.1.1	Cloud formation	0,00	1,00
39	94	50.4.1.2	Cloud types and cloud classification	1,00	1,00
40	119	50.10.1.5	Aircraft observations and reporting	0,00	1,00
Total:28%				11,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Asaba, Joshua

Location: Uganda Civil Aviation Authority

Your Result: 11,00 from 40,00 Points (27,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
50. METEOROLOGY	40	11,00	40,00	27,50
50.2. WIND	2	2,00	2,00	100,00
50.3. THERMODYNAMICS	1	0,00	1,00	0,00
50.4. CLOUDS AND FOG	2	1,00	2,00	50,00
50.10. METEOROLOGICAL INFORMATION	1	0,00	1,00	0,00
Total	40	11,00	40,00	27,50

1

What type turbulence should be reported when it causes changes in altitude and/or attitude more than two-thirds of the time, with the aircraft remaining in positive control at all times?

☐ Continuous severe chop.

☒ Intermittent moderate turbulence.

☐ Continuous moderate turbulence.

2

What determines the structure or type of clouds which will form as a result of air being forced to ascend?

- ☒ The method by which the air is lifted.
- ☐ The relative humidity of the air after lifting occurs.
- ☐ The stability of the air before lifting occurs.

3

The section of the Aviation Area Forecast (FA) entitled VFR Clouds and Weather contains a summary of

- ☐ sky condition, cloud heights, visibility, obstructions to vision, precipitation, and sustained surface winds of 20 knots or greater.
- ☐ only those weather systems producing liquid or frozen precipitation, fog, thunderstorms, or IFR ceilings.
- ☒ forecast sky cover, cloud tops, visibility, and obstructions to vision along specific routes.

4

A common type of ground or surface based temperature inversion is that which is produced by

- ☐ ground radiation on clear, cool nights when the wind is light.
- ☒ the movement of colder air over warm air, or the movement of warm air under cold air.
- ☐ warm air being lifted rapidly aloft in the vicinity of mountainous terrain.

5

Which weather phenomenon signals the beginning of the mature stage of a thunderstorm?

☐ Growth rate of cloud is maximum.

☐ The appearance of an anvil top.

☒ The start of rain.

6

Which primary source contains information regarding the expected weather at the destination airport, at the ETA?

☐ Radar Summary and Weather Depiction Charts.

☒ Terminal Aerodrome Forecast.

☐ Low-Level Prog Chart.

7

Which weather condition is an example of a nonfrontal instability band?

☒ Frontogenesis.

☐ Advective fog.

☐ Squall line.

8

If a SIGMET alert is announced, how can information contained in the SIGMET be obtained?

- ☒ **ATC will announce the hazard and advise when information will be provided in the Flight control center broadcast.**
- ☐ **By contacting the nearest Flight control center**
- ☐ **By contacting a weather watch station.**

9

What minimum weather conditions must be forecast for your ETA at an alternate airport, that has only a VOR approach with standard alternate minimums, for the airport to be listed as an alternate on the IFR flight plan?

- ☐ 800-foot ceiling and 2 statute miles visibility.
- ☒ 1,000-foot ceiling and visibility to allow descent from minimum en route altitude (MEA), approach, and landing under basic VFR.
- ☐ 800-foot ceiling and 1 statute mile visibility.

10

What type turbulence should be reported when it causes slight, rapid, and somewhat rhythmic bumpiness without appreciable changes in attitude or altitude, less than one-third of the time?

- ☐ Moderate chop.
- ☒ Moderate turbulence.
- ☐ Occasional light chop.

11

What is the wind shear forecast in the following TAF?
TAF|KCVG 231051Z 231212
12012KT 4SM -RA BR OVC008|WS005/27050KT TEMPO 1719 1/2SM -RA FG|FM1930
09012KT 1SM -DZ BR VV003 BECMG 2021 5SM HZ=

☐ 50 feet AGL from 270 degrees at 50 KT.

☐ 500 feet AGL from 270 degrees at 50 KT.

☒ 5 feet AGL from 270 degrees at 50 KT.

12

What relationship exists between the winds at 2,000 feet above the surface and the surface winds?

- ☐ The winds at 2,000 feet and the surface winds flow in the same direction, but the surface winds are weaker due to friction.
- ☒ The surface winds tend to veer to the right of the winds at 2,000 feet and are usually weaker.
- ☐ The winds at 2,000 feet tend to parallel the isobars while the surface winds cross the isobars at an angle toward lower pressure and are weaker.

13

Which is an effect of ice, snow, or frost formation on an airplane?

- ☒ Increased stall speed.
- ☐ Increased angle of attack for stalls.
- ☐ Increased pitchdown tendencies.

14

The visibility entry in a Terminal Aerodrome Forecast (TAF) of P6SM implies that the prevailing visibility is expected to be greater than

☐ 6 statute miles.

☒ 6 nautical miles.

☐ 6 kilometers.

15

When flying over the top of a severe thunderstorm, the cloud should be overflown by at least

☒ 2,500 feet.

☐ 500 feet above any moderate to severe turbulence layer.

☐ 1,000 feet for each 10 knots windspeed.

16

The conditions necessary for the formation of stratiform clouds are a lifting action and

- ☒ stable, moist air.
- ☐ unstable, dry air.
- ☐ unstable, moist air.

17

Absence of the sky condition and visibility on an ATIS broadcast indicates that

- ☒ the sky condition is clear and visibility is unrestricted.
- ☐ the ceiling is at least 5,000 feet and visibility is 5 miles or more.
- ☐ weather conditions are at or above VFR minimums.

18

Convective currents are most active on warm summer afternoons when winds are

☒ **light.**

☐ **strong.**

☐ **moderate.**

19

Which is true regarding ridge soaring with the wind direction perpendicular to the ridge?

- ☐ When very close to the surface of the ridge, the glider's speed should be reduced to the minimum sink speed.
- ☐ If the glider drifts downwind from the ridge and sinks slightly lower than the crest of the ridge, the glider should be turned away from the ridge and a high speed attained.
- ☒ When the wind and lift are very strong on the windward side of the ridge, a weak sink condition will exist on the leeward side.

20

In which situation is advection fog most likely to form?

- ☐ A light breeze blowing colder air out to sea.
- ☒ Warm, moist air settling over a warmer surface under no-wind conditions.
- ☐ An air mass moving inland from the coast in winter.

21

(Refer to Figure 9.) The Severe Weather Outlook Chart depicts

See Attachments:

1. cfi_9.jpg

cfi_9.jpg

(Attachment
No.1)

- ☐ areas of probable severe thunderstorms by the use of single hatched areas on the chart.
- ☒ areas of general thunderstorm activity (excluding severe) by the use of hatching on the chart.
- ☐ areas of forecast, severe or extreme turbulence, and areas of severe icing for the next 24 hours.

22

What weather feature occurs at altitude levels near the tropopause?

- ☐ Maximum winds and narrow wind shear zones.
- ☒ Thin layers of cirrus (ice crystal) clouds at the tropopause level.
- ☐ Abrupt temperature increase above the tropopause.

23

What values are used for Winds Aloft Forecasts?

- ☒ True direction and knots.
- ☐ True direction and MPH.
- ☐ Magnetic direction and knots.

24

Area forecasts generally include a forecast period of 18 hours and cover a geographical

- ☐ area the size of several states.
- ☒ area less than 3000 square miles.
- ☐ terminal area.

25

When are ATIS broadcasts updated?

- ☐ Only when the ceiling and/or visibility changes by a reportable value.
- ☐ Upon receipt of any official weather, regardless of content change or reported values.
- ☒ Every 30 minutes if weather conditions are below basic VFR; otherwise, hourly.

26

What is meant by the Special METAR weather observation for KBOI?
091854Z 32005KT 1 1/2SM RA BR OVC007 17/16 A2990 RMK RAB12

☐ Rain and fog obscuring two-tenths of the sky; rain began at 1912Z.

☒ Rain and overcast at 1200 feet AGL.

☐ Rain and mist obstructing visibility; rain began at 1812Z.

27

If a SIGMET alert is announced, how can information contained in the SIGMET be obtained?

- ☐ By contacting the nearest AFSS.
- ☐ By contacting a weather watch station.
- ☒ ATC will announce the hazard and advise when information will be provided in the FSS broadcast.

28

At what time are current AIRMETs broadcast in their entirety by the Hazardous Inflight Weather Advisory Service (HIWAS)?

- ☐ There is a continuous broadcast over selected VORs of Inflight Weather Advisories.
- ☐ Every 15 minutes until the AIRMET is canceled.
- ☒ 15 and 45 minutes after the hour during the first hour after issuance, and upon receipt.

29

What type turbulence should be reported when it momentarily causes slight, erratic changes in altitude and/or attitude, one-third to two-thirds of the time?

- ☐ Moderate chop.
- ☐ Occasional light chop.
- ☒ Intermittent light turbulence.

30

Which statement pertaining to the following Terminal Aerodrome Forecast (TAF) is true?
|TAF|KMEM 091135Z 0915 15005KT 5SM HZ BKN060|FM1600 VRB04KT P6SM SKC

- ☐ Wind in the valid period implies surface winds are forecast to be greater than 5 KTS.
- ☐ SKC in the valid period indicates no significant weather and sky clear.
- ☒ Wind direction is from 160__degrees__ at 4 KTS and reported visibility is 6 statute miles.

31

(Refer to Figure 152.) What weather conditions are depicted in the area indicated by arrow B on the Radar Summary Chart?

See Attachments:

1. atp_152.jpg

figure 152

(Attachment
No.2)

- ☒ Weak echoes; heavy rain showers; area movement toward the southeast.
- ☐ Strong echoes; moderate rain showers; no cell movement.
- ☐ Weak to moderate echoes; rain showers increasing in intensity.

32

In what localities is advection fog most likely to occur?

- ☒ Coastal areas.
- ☐ Level inland areas.
- ☒ Mountain slopes.

33

Which family of clouds is least likely to contribute to structural icing on an aircraft?

☐ Low clouds.

☒ Clouds with extensive vertical development.

☐ High clouds.

34

What lifts advection fog into low stratus clouds?

- ☐ Dryness of the underlying land mass.
- ☒ Surface winds of approximately 15 knots or stronger.
- ☐ Nighttime cooling.

35

Which forces are balanced in case of geostrophic winds?

- ☐ Friction force, pressure gradient force, Coriolis force.
- ☐ Pressure gradient force, centrifugal force, friction force.
- ☐ Pressure gradient force, Coriolis force, centrifugal force.
- ☒ Pressure gradient force, Coriolis force.

36

A high pressure area (slack pressure gradient) covers part of the Mediterranean Sea and coastal region during the summer. What surface wind direction is likely at an airport at the coast on a sunny afternoon?

- ☒ Sea to land.
- ☐ Variable.
- ☐ Parallel to the coastline.
- ☐ Land to sea.

37

When advection fog has developed, what may tend to dissipate or lift the fog into low stratus clouds?

☐ Wind stronger than 15 knots.

☒ Temperature inversion.

☐ Surface radiation.

38

Which of the following phenomena is least likely to lead to the formation of a Cumulonimbus with thunderstorm ?

- ☐ Convection
- ☐ Orographic lifting
- ☒ Convergence
- ☐ Ground radiation

Candidate: Joshua Asaba
Date of birth: 1990-02-24

Subject: 050-Meteorology
Date: 2014-07-18

39

Which of the following cloud types is found at high levels?

☒ CI

☐ AS

☐ CU

☐ SC

40

A special air-report

- ☒ may trigger a SIGMET message
- ☐ may be the reason for closing a part of an airway due to meteorological conditions
- ☒ is used to produce better aerodrome forecasts (TAF)
- ☐ may be the reason for closing an airport due to meteorological conditions

Attachment No. 1

Questions 21

cfi_9.jpg

cfi_9.jpg

1.



2.



3.



Figure 9. Surface Analysis Chart Symbols © ASA

Attachment No. 2

Questions 31

figure 152

atp_152.jpg

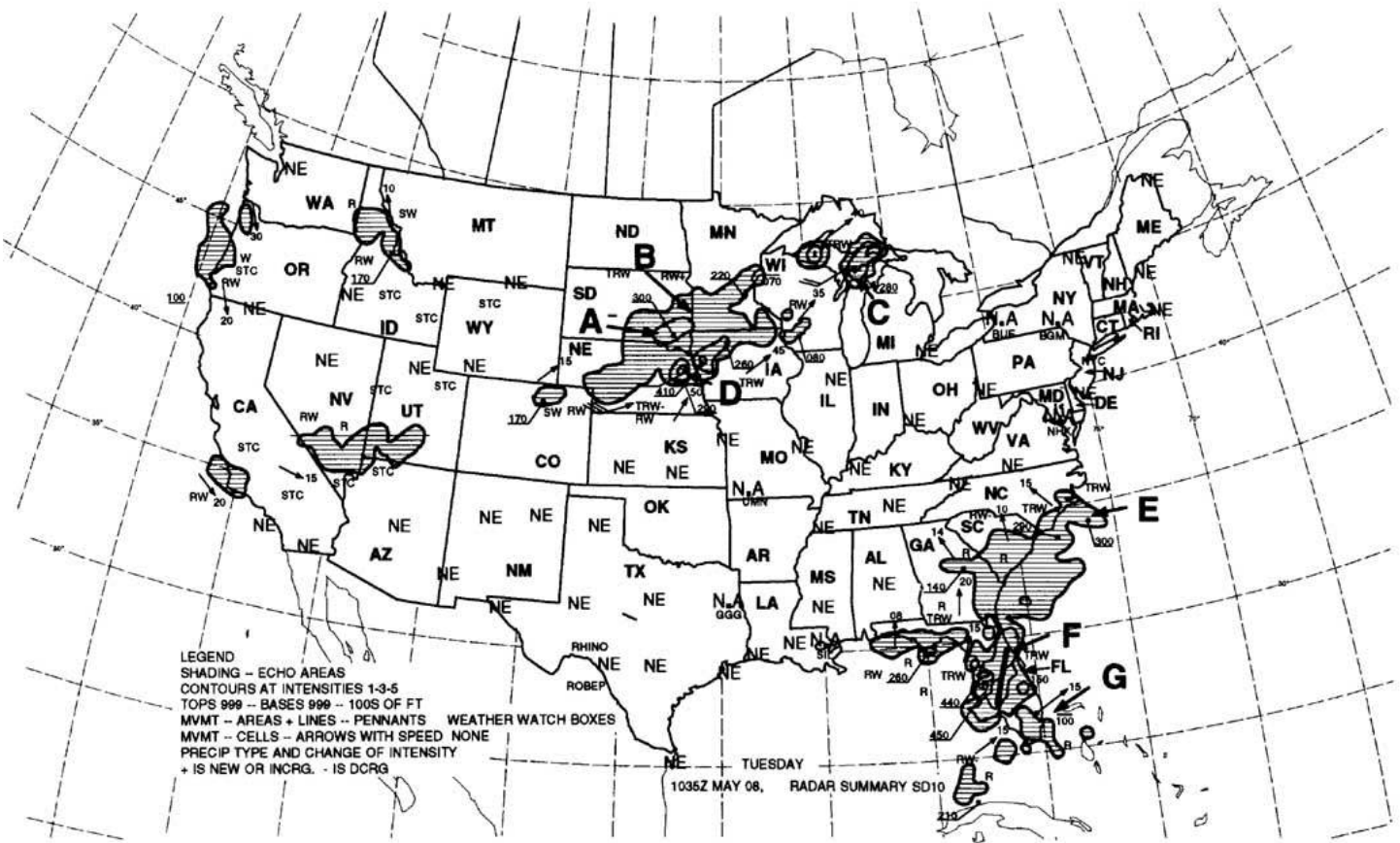


Figure 152. Radar Summary Chart © ASA

